

DATABASE AND USERS

INTRODUCTION

- Databases and Database system have become an essential component of everyday life in our society.
- They play a critical role in almost all the areas (business, electronic commerce , engineering , medicine , law , education etc)
- **Applications** : Multimedia databases, Geographic Information System (GIS), Data warehouses, Online analytical processing, Real time and active database technology.

DATABASE TERMINOLOGIES

- **Data**: It is a known facts that can be recorded either manually electronically and have implicit meaning.
Eg: Name, telephone numbers, place, address
- Data is primarily classified into two types
 Numeric and Alphanumeric
- **Datum**: is a singular form of data (single piece of information)
- **Automation of data**: keeping data in electronic device

Different options to store data

1. **Storing data in variable:** it will be stored in primary storage RAM. It is volatile in nature
2. **Storing in file system:** it is collection of related data stored in secondary storage-hard disk

Disadvantages of file system

1. Data redundancy (Duplication of data) and inconsistency

The same data may be stored in several files. This redundancy leads to higher storage. Since all these files are independent to each other, it may lead to data inconsistency that is various copies of the same data in different files may not match.

Eg: A student record might be stored in college office, library and in examination department. When any changes are required in student details then it has to be updated in all places otherwise, it will create a risk of outdated values of data.

2. Accessing Anomalies (Difficulty in accessing data)

Accessing anomalies means it is not easy to access data in desired way. The conventional file processing system do not allow data to retrieved in a convenient and efficient manner according to user choice.

Eg: If HOD wants those student details having attendance shortage in two or more subjects or who have secured more than 75% in examination, then we require to create separate programs for each of these requirement.

3. Poor Data Integrity

A collection of data is integrated only if it meets certain consistency constraints (data validation). A programmer puts these constraints in the programs by adding appropriate code. In file system poor data integrity arises because it is very difficult to add new constraint each time when new constraints are to be added.

Eg: A maximum marks for subject can never be more than 100.

4. Poor data security

There is no security provided in file system to secure the data from unauthorized access. Anyone can easily modify and change the data stored in the files.

Eg: If student can access his data in the library, then he can easily change book issued date and also he can change fine details to zero.

All the users must have some restriction of accessing data up to a certain level.

5. Atomicity Problem

Atomicity means, system should ensures that either all the steps in transaction are executed successfully or none of them. If any step fails, the system rollback the entire transaction to ensure that the database remains consistent state. It is difficult to ensure atomicity in a file processing system when transaction failure occurs due to power failure, networking problems etc.

Eg: if you are buying a ticket from railway and you are in the process of money transaction. Suddenly your internet got disconnected then you may or may not have paid for the ticket. If you have paid then your ticket has to be booked and if not, nothing should be charged to the passenger.

6. Data Isolation

In file system data are scattered in various files and it may be in different formats. Writing program to retrieve the appropriate data is difficult.

7. Concurrent access

In the file processing system it is not possible to access a same file for different transactions at same time.

DATABASE TERMINOLOGIES

- **Record**: It is a collection of related data items
- **Information**: It is a processed form of data on which decisions are based and provides meaningful values.
Eg: “The age of Ravi is 25”
- **Database**: It is an organized collection of related information.
- **Knowledge**: It is practical use of information.

In order to store and manage the database we require **DBMS**

Implicit Properties of the database:

1. A database represents some aspects of the real world , sometimes called the mini-world. Changes to the mini-world are reflected in the data base .
2. A database is a logically coherent collection of data with some inherent meaning.
3. A data base is designed , built and populated with data for a specific purpose . It has an intended group of users and some preconceived applications in which these users are interested.

DATABASE MANAGEMENT SYSTEM (DBMS)

DBMS: It is a program which provides the user with the facility to define, to store, to retrieve, to manage, to modify and to control the access to the data in the database.

Eg: MySQL, Oracle, SQL Server, IBM DB2 etc are a very popular commercial DBMS which is used in different applications.

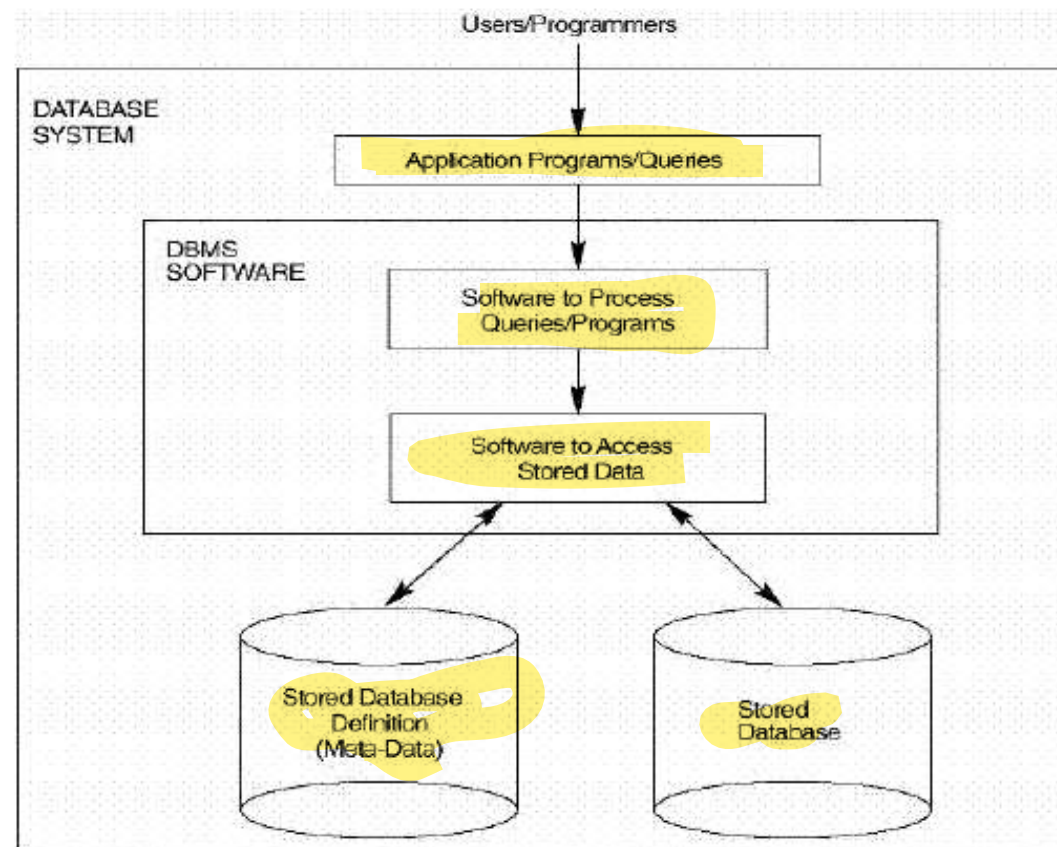
Facilities of DBMS

- 1. Defining** - Specification of the data types , structures, and constraints.
- 2. Constructing** - The process of storing the data itself on some media.
- 3. Manipulating** - Includes functions such as querying, updating and generating reports.
- 4. Sharing** - Concurrently accessing the database by multiple users and applications.

- An **application program** accesses the database by sending queries or requests for data to the DBMS.
- A **query** typically causes some data to be retrieved.
- A **transaction** may cause some data to be read and some data to be written into the database.
- Functions provided by the database:
 - **Protection:**
 - **System Protection:** protection against hardware and software malfunctions.
 - **Security Protection:** against unauthorized or malicious access.
 - **Maintenance:** allowing the system to evolve as the requirements change over time.

Database system: Database + DBMS software

DATABASE SYSTEM



A simplified Database System Environment

CHARACTERISTICS OF THE DATABASE APPROACH

1. Self-describing nature of a database system
2. Insulation between programs and data and Data Abstraction
3. Support of multiple views of the data
4. Sharing of data and multi-user transaction processing

SELF-DESCRIBING NATURE OF A DATABASE SYSTEM:

- A DBMS catalog stores the description of a particular database (e.g. data structures, types, and constraints)
- The information stored in the catalog is called **meta-data**.
- This allows the DBMS software to work with different database applications.

INSULATION BETWEEN PROGRAMS AND DATA, DATA ABSTRACTION

- Called **program-data independence**.
- Allows changing data structures and storage organization without having to change the DBMS access programs.
- Data Abstraction:
 - A **data model** is used to hide storage details and present the users with a conceptual view of the database.
 - Programs refer to the data model constructs rather than data storage details.

SUPPORT OF MULTIPLE VIEWS OF THE DATA:

- Each user may see a different view of the database, which describes **only** the data of interest to that user.
- A view may be a subset of the database or it may contain virtual data that is derived from the database files but is not explicitly stored.

SHARING OF DATA AND MULTI-USER TRANSACTION PROCESSING:

- Allowing a set of **concurrent users** to retrieve from and to update the database.
- *Concurrency control* within the DBMS guarantees that each **transaction** is correctly executed or aborted
- *Recovery* subsystem ensures each completed transaction has its effect permanently recorded in the database

Database Users

Users may be divided into:

- Those who actually use and control the database content, and those who design, develop and maintain database applications called **Actors on the scene**.
- Those who design and develop the DBMS software and related tools and the computer systems operators called **Workers behind the scene**.

DATABASE ADMINISTRATORS (DBA)

- Responsible for authorizing access to the database, for coordinating and monitoring its use, acquiring software and hardware resources.
- Accountable for problems such as breach of security or poor system response time.

DATABASE DESIGNERS

- Responsible for identifying the data to be stored in the database and for choosing appropriate structures to represent and store the data.
- It is the responsibility of the database designers to communicate with the end-users and understand their requirements.

END USERS

- These are the people whose jobs require access to the database for querying, updating, and generating reports.

- Several categories of end-users:
 1. Casual End users
 2. Naive or parametric End users
 3. Sophisticated End users
 4. Standalone users

1. **Casual End users:** occasionally access the database, but they might require different information each time.
2. **Naive or parametric end users:** Their main job revolves around constantly querying and updating the database, using standard types of queries and updates called **canned transactions** that have been carefully programmed and tested.
3. **Sophisticated end users:** includes engineers, scientists, business analysts and others who are thoroughly familiar with the DBMS.
4. **Standalone users:** maintains personal databases by using ready-made program packages that provide easy-to-use menu-based or graphics-based interfaces.

Workers behind the scene

- **DBMS system designers and implementers:** design and implement the DBMS modules and interfaces as a software package.
- **Tool developers:** design and implement tools-the software packages that facilitate database modeling and design, database system design and improved performance.
- **Operators and maintenance personnel:** are responsible for the actual running and maintenance of the hardware and software environment for the database system.